



TAS LAB

Trustworthy AI and
Autonomous Systems Lab

THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

Department of
Aeronautical and Aviation Engineering
航空及民航工程學系

Remarkable Team Milestones in 2024

- PolyU-Wuxi ITS & Unmanned Systems Center launched
- RGC Early Career Scheme awarded; Associate Editor, IEEE T-VT
- Hosted IPIN 2024 in Hong Kong; drones featured by TVB, RTHK, Ming Pao

2024 TASLAB Annals

Webpage: <https://polyu-taslab.github.io>

Trustworthy AI and Autonomous Systems Lab, PolyU

17

journal papers

9

conference papers

24

members & visitors

728

Google Scholar citations

6

awards & recognitions

30

news stories on our website

Faculty & Staff



Dr Weisong Wen
Asst Prof, AAE



Dr Runqiu Yang
Postdoc



Dr Shiyu Bai
Postdoc



Mr Pei Ma
RA



Ms Akida Tursun
Staff, PolyU-Wuxi



Mr Yun Wang
RA, PolyU-Wuxi

PhD/MPhil Student Members (and Undergraduates)



Xi ZHENG
PhD



Xikun LIU
PhD



Yihan ZHONG
PhD



Runzhi HU
PhD



Feng HUANG
PhD



Peiwen YANG
PhD (2023&€)



Yifeng CAO
PhD (2023&€)



Ruijie XU
PhD (2023&€)



Yuanyuan ZHANG
PhD (2023&€)



Naigui XIAO
PhD (2024&€)



Bingxian LI
PhD (2024&€)



Ziqi ZHANG
PhD (2024&€)



Yixin GAO
PhD (2024&€)



Jiaqi ZHAO
MPhil (2024&€)



Yingming CHEN
MPhil (2024&€)



Shaoting QIU
MPhil (2024&€)



Yuling CHENG
MPhil (2024&€)



Qijun QIN
UG, URIS

2024 in Review

In 2024 the lab grew to 24 members and opened its Mainland base, the PolyU-Wuxi ITS & Unmanned Systems Center. The team was awarded an RGC Early Career Scheme project on LiDAR-aided GNSS, hosted the IPIN 2024 conference in Hong Kong and chaired the "Intelligent Vehicle Meets Urban" workshop at ITSC 2024 in Edmonton. Dr Wen became Associate Editor of IEEE T-VT and was again listed among the Top 2% Scientists. 17 journal and 9 conference papers appeared, and our window-cleaning and aerial 3D-printing drones were featured by TVB, RTHK and Ming Pao.

Selected papers

1. X. Xia, W. Wen, L.-T. Hsu. Integrity-constrained factor graph optimization for GNSS positioning in urban canyons. NAVIGATION
2. S. Bai, W. Wen, L.-T. Hsu, P. Yang. Factor graph optimization-based smartphone IMU-only indoor SLAM with multihypothesis turning-behavior loop closures. IEEE TAES
3. Y. Zhong, W. Wen, L.-T. Hsu. Trajectory smoothing using GNSS/PDR integration via factor graph optimization. IEEE IoT-J
4. R. Xu, S. Chen, S. Bai, W. Wen. Nonlinearity-aware ZUPT-aided pedestrian inertial navigation based on cubature Kalman filter. IEEE TIM

Research & Innovation Awards

World's Top 2% Scientists

Stanford University / Elsevier · second year

Best Student Paper Award, ION GNSS+ 2024

Co-supervised student Penggao Yan · Jackknife test for GNSS fault detection

Most-cited paper of 2023, NAVIGATION

FGO for GNSS/INS integration, recognised at ION GNSS+ 2024

Associate Editor, IEEE T-VT

IEEE Transactions on Vehicular Technology (2024-)

Third Prize, 10th HK University Student I&E Competition

UG team incl. Ziqi Zhang



Merit Award, Best URIS Research Project
Qijun Qin (UG)

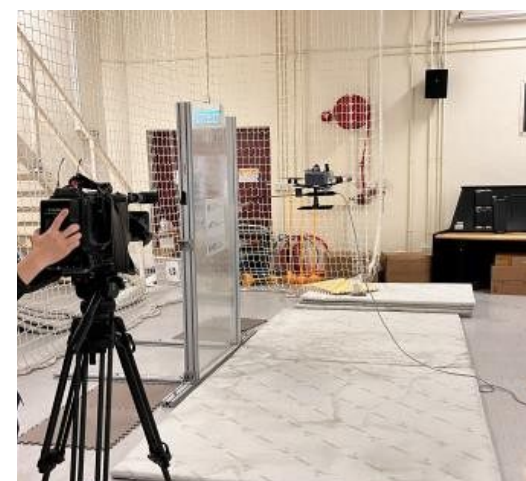
International Journal & Conference Publications

IEEE T-IV IEEE Transactions on Intelligent Vehicles (6)	IEEE TIM IEEE Transactions on Instrumentation (3)	IEEE IoT-J IEEE Internet of Things Journal (2)	JOURNAL MST Measurement Science and Technology (2)	IEEE TAES IEEE Transactions on Aerospace and Electronic Systems (1)	JOURNAL IET-ITS IET Intelligent Transport Systems (1)	JOURNAL JDSIS Journal of Data Science and Intelligent Systems (1)	ION NAVI NAVIGATION, Journal of the ION (1)
CONF. ICRA Conference papers (2)	CONF. ION GNSS+ Conference papers (2)	CONF. ICUAS Conference paper (1)	CONF. ION ITM Conference paper (1)	CONF. IPIN Conference paper (1)	CONF. ISPRS Conference paper (1)	CONF. ITSC Conference paper (1)	

Talks, Visits & Service

- Mar** Hunan University · Changsha
- Apr** Hohai University · Nanjing
- Apr** Westwell · Shanghai
- Apr** PolyU FCE sharing · Hong Kong
- Jun** LegCo Member visits TAS Lab · PolyU
- Jun** Tsinghua University seminar · Beijing
- Jun** World Intelligence Expo · Tianjin
- Jul** Wuhan University · Wuhan
- Sep** Session chair (UAVs) & workshop chair, ITSC 2024 · Edmonton
- Nov** ITS Congress: HD maps for smart roads · Hangzhou
- Dec** Lanzhou University: 3D LiDAR aided GNSS · Lanzhou
- Dec** UM-HNU-PolyU academic salon on autonomous systems · Macau
- Dec** Dr Runqiu Yang at HKPC Academy · Hong Kong

Media Publicity



TVB Finance: Hong Kong's low-altitude economy (May)



Ming Pao: drones for building cleaning



RTHK interview on drones and the LAE (Jun)



TVB interview with TAS Lab members

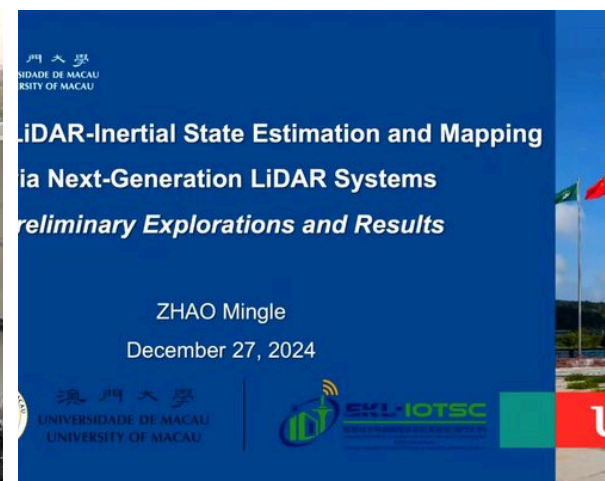
Group Gathering / Impact Story



IPIN 2024, hosted in Hong Kong



PolyU-Wuxi ITS & Unmanned Systems Center



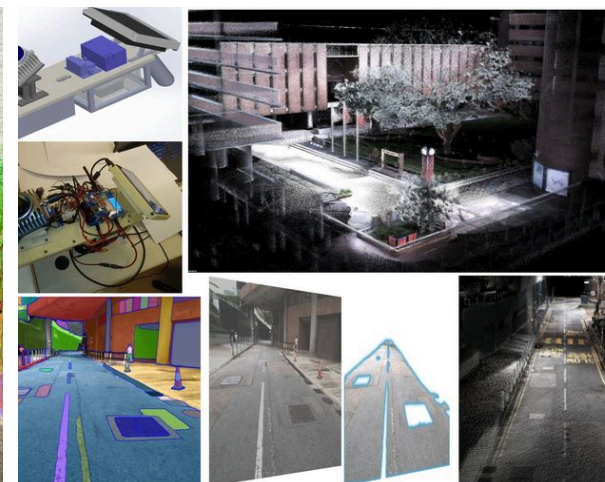
UM-HNU-PolyU salon (Dec)



Dr Runqiu Yang at HKPC Academy (Dec)



Field study in Sha Tau Kok (Dec)



URIS project demonstration